Immigration Narratives vs. Enforcement: A Total Survey Error Approach

Sagnik Chakravarty

Abstract

This study evaluates the relationship between immigration-related narratives and enforcement activity using a Total Survey Error (TSE) framework. Combining news coverage, social media sentiment, and administrative enforcement data, I assess whether narrative indicators track underlying enforcement trends. The results show only modest contemporaneous association between narrative and enforcement series. News coverage is moderately correlated with CBP encounters ($r = 0.32$), while Reddit sentiment shows a somewhat stronger contemporaneous association ($r = 0.42$) over a shorter overlap period. A lagged correlation analysis indicates that news coverage is more strongly associated with next-period CBP encounters ($r = 0.48$) than the reverse ($r = 0.27$), suggesting that narrative and enforcement series may move with delay rather than in lockstep. A simple regression of standardized CBP encounters on standardized news coverage yields limited explanatory power ($R^2 \approx 0.10$). Influence diagnostics show that this weak positive relationship is not driven by a single outlier. Overall, the findings indicate that narrative indicators are partially related to enforcement activity but do not provide a strong or consistent stand-alone proxy for underlying enforcement conditions.

Introduction

Immigration discourse in the United States is heavily shaped by media coverage and online platforms. These narrative systems influence public perception, policy debates, and political behavior. However, whether such narratives accurately reflect underlying enforcement dynamics remains an open empirical question. This matters because researchers, journalists, and policymakers often use narrative indicators—such as media attention, online sentiment, or issue salience—as shorthand for broader immigration conditions.

From a Total Survey Error (TSE) perspective, narrative data can be conceptualized as a measurement process rather than a direct representation of reality. Unlike survey instruments

or administrative reporting systems, news and social discourse are not designed to produce unbiased estimates of social conditions. Instead, they are shaped by editorial selection, platform incentives, engagement dynamics, and framing effects. For that reason, they may align only imperfectly with the underlying processes they appear to measure.

This paper examines the relationship between narrative salience and enforcement activity by integrating multiple monthly data sources: (1) news coverage from GDELT, (2) social media sentiment from Reddit, and (3) administrative enforcement data from CBP and ICE. The central question is whether narrative indicators track enforcement trends or instead show limited, delayed, or unstable alignment.

The contribution of the paper is methodological as well as substantive. Substantively, it assesses whether immigration narratives move with enforcement activity. Methodologically, it applies a TSE framework to computational social science data by asking whether narrative measures behave like valid indicators of the underlying construct or whether they should instead be treated as incomplete and potentially lagged signals.

Methods

Data Sources. Monthly time series were constructed from:

- news coverage from GDELT, including article volume and average tone,
- Reddit-based sentiment measures,
- enforcement data from CBP encounters and ICE arrests.

The analysis is conducted at the monthly level in order to align narrative and enforcement series on a common time scale.

Standardization. To ensure comparability across variables with different units and scales, all major series were transformed into standardized z-scores:

$$z_{it} = \frac{x_{it} - \mu_i}{\sigma_i}$$

where x_{it} is the value of variable i at time t, and μ_i and σ_i are the mean and standard deviation of series i.

Divergence Measure. To quantify mismatch between narratives and enforcement, I define:

$$D_t = z_t^{\text{news}} - z_t^{\text{CBP}}$$

Positive values indicate that narrative salience exceeds the CBP benchmark in standardized terms, while negative values indicate that the benchmark exceeds narrative salience.

Correlation Analysis. To evaluate linear association, I compute contemporaneous correlations between:

$$\mathrm{Corr}(z_t^{\mathrm{news}}, z_t^{\mathrm{CBP}}) \qquad \text{and} \qquad \mathrm{Corr}(z_t^{\mathrm{sentiment}}, z_t^{\mathrm{CBP}}).$$

Because the professor's feedback raised the possibility of lagged movement rather than simple contemporaneous misalignment, I also examine one-period lagged correlations:

$$\mathrm{Corr}(z_t^{\mathrm{news}}, z_{t-1}^{\mathrm{CBP}}) \qquad \text{and} \qquad \mathrm{Corr}(z_{t-1}^{\mathrm{news}}, z_t^{\mathrm{CBP}}).$$

Regression Model. I estimate a linear model of the form:

$$z_t^{\mathrm{CBP}} = \beta_0 + \beta_1 z_t^{\mathrm{news}} + \epsilon_t$$

where ϵ_t captures unexplained variation.

Residual Analysis. Model residuals are defined as:

$$\hat{\epsilon}_t = z_t^{\mathrm{CBP}} - \hat{z}_t^{\mathrm{CBP}}.$$

Residual patterns are examined descriptively to assess whether there are extended periods of under- or over-prediction.

Influence Diagnostics. To address the possibility that the estimated regression slope is driven by one highly influential observation, I examine Cook's distance. In the fitted model, the largest Cook's distance is approximately 0.08, which is small and suggests that the weak positive relationship is not being driven by a single outlier.

Results

Descriptive overview. Summary statistics for the main narrative and enforcement indicators are reported in Table 1 in the Appendix. These statistics show that the series differ substantially in scale and temporal coverage, which motivates the use of standardized comparisons.

Standardized trends. Figure 1 presents standardized trends across narrative and enforcement indicators. Some co-movement is visible, but the series do not move together consistently over time. In particular, there are several periods in which narrative indicators

rise or remain elevated while enforcement indicators decline or remain stable. At the same time, the figure also suggests that outright independence would be too strong a conclusion, since the series do occasionally move in the same direction.

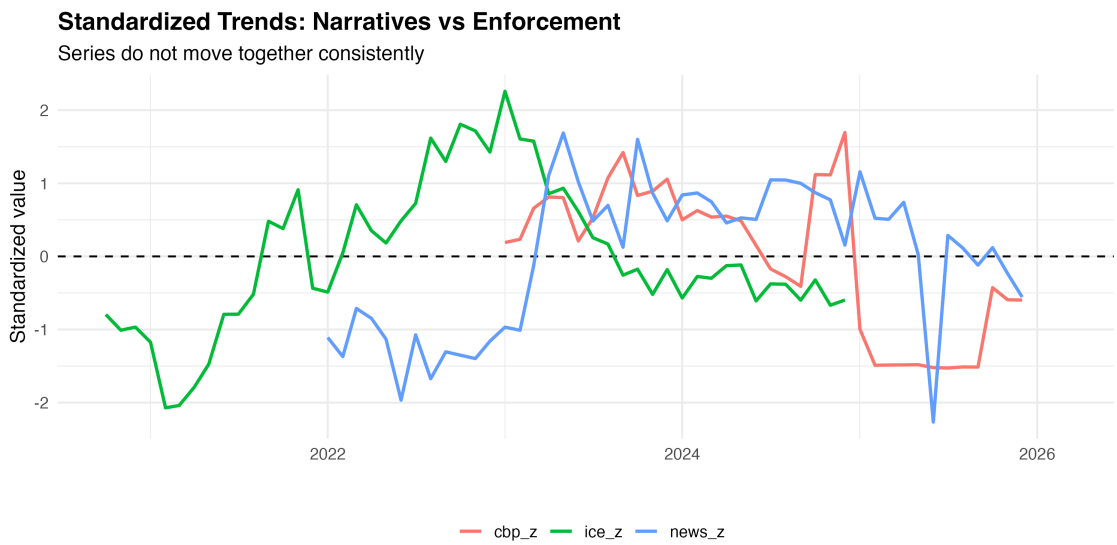

Figure 1: Standardized Trends: Narratives vs. Enforcement

Contemporaneous association. The correlation between standardized news coverage and standardized CBP encounters is $r = 0.32$, indicating a modest positive association. The correlation between standardized Reddit sentiment and standardized CBP encounters is $r = 0.42$, which is somewhat stronger, though it is based on a shorter overlap period and should therefore be interpreted more cautiously. Taken together, these estimates suggest that narrative indicators are not unrelated to enforcement activity, but the contemporaneous relationship is limited.

Lagged association. A one-period lag analysis suggests that timing may matter. The correlation between current news coverage and lagged CBP encounters is $r = 0.48$, whereas the correlation between lagged news coverage and current CBP encounters is $r = 0.27$. This pattern suggests that narrative and enforcement series may move with delay, and that some of the apparent mismatch in contemporaneous plots may reflect timing differences rather than complete disconnection. Because this is a simple lagged correlation rather than a full time-series model, it should be interpreted as suggestive rather than definitive.

Divergence. Figure 2 plots the divergence measure

$$D_t = z_t^{\text{news}} - z_t^{\text{CBP}}.$$

The divergence series exhibits sustained positive and negative stretches rather than short-

lived random oscillation. However, on the basis of the statistical evidence above, these deviations are better interpreted as partial and time-varying misalignment than as direct proof of systematic measurement bias. In other words, the divergence plot shows that the two series often differ in level and timing, but it does not by itself establish the source of that difference.

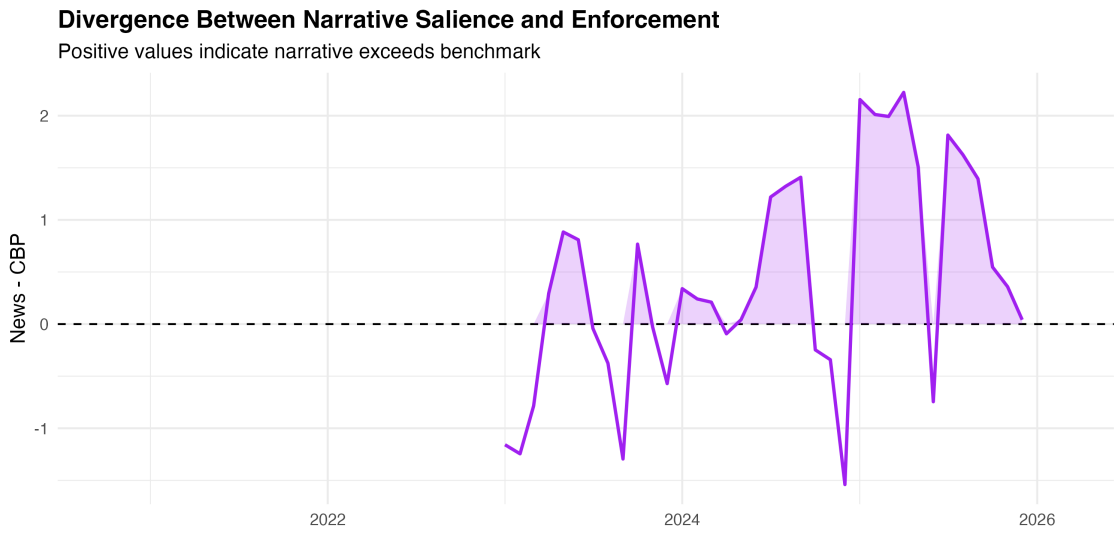

Figure 2: Divergence Between Narrative Salience and Enforcement

Regression relationship. Figure 3 presents the regression relationship between news indicators and CBP encounters. The estimated $R^2 \approx 0.10$ indicates that narrative variables explain only a small fraction of variation in enforcement outcomes. The slope is positive but weak, which is consistent with the modest contemporaneous correlation reported above.

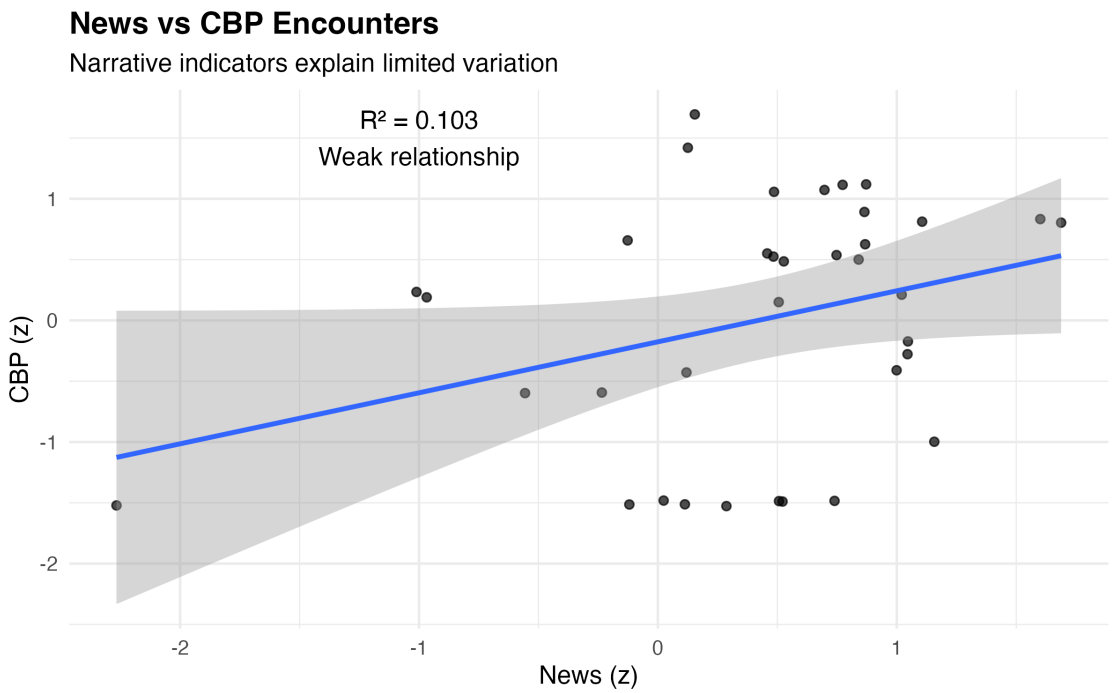

Figure 3: News vs. CBP Encounters

Influence and robustness. Cook's distance diagnostics indicate that the regression slope is not being forced by a single extreme observation. The largest Cook's distance is approximately 0.08, which is small in practical terms. This matters because it strengthens the interpretation that the weak positive relationship is a property of the full sample rather than an artifact of one outlying point.

Residual patterns. Figure 4 in the Appendix shows residuals from the news–CBP regression over time. The residual series shows periods of repeated under- and over-prediction. I interpret this cautiously: the residual pattern indicates that a simple contemporaneous linear model does not capture the full relationship between narratives and enforcement. It does not, by itself, prove systematic bias. Instead, it suggests that the relationship may be nonlinear, lagged, or influenced by omitted dynamics.

Interpretation. Overall, the results support a more limited conclusion than the original draft. Narrative indicators are partially aligned with enforcement activity, but they do not provide a strong or consistent contemporaneous proxy for it. The lagged correlation results suggest that some apparent misalignment may reflect timing differences rather than complete disconnection. Thus, the strongest defensible conclusion is that narrative indicators are informative but incomplete measures of enforcement-related conditions.

Discussion

Within a TSE framework, these findings suggest caution in how narrative indicators are used. The results do not justify the strongest version of the claim that narratives are simply "biased measurement systems" in a direct statistical sense. Instead, the evidence suggests that narratives are best understood as partial, mediated, and temporally uneven indicators.

Several mechanisms could produce this pattern.

- **Coverage differences:** News organizations select stories according to editorial priorities and event salience rather than according to a sampling design.
- **Framing differences:** The same enforcement environment may be interpreted differently across outlets and platforms, producing variation in tone and attention even when underlying conditions are similar.
- **Temporal misalignment:** Narrative indicators may respond more quickly or more slowly than administrative indicators, producing the lagged pattern observed here.
- **Platform amplification:** Social media may intensify selected signals through engagement dynamics, making narrative attention more volatile than the benchmark processes themselves.

These mechanisms have real-world implications. If policymakers, journalists, or researchers interpret spikes in immigration coverage as direct evidence of increased enforcement, they may overreact to attention cycles rather than to underlying administrative conditions. Conversely, if enforcement changes occur without receiving comparable narrative attention, important shifts in policy or practice may be underestimated in public discourse. For survey researchers, the implication is also important: respondents exposed to different media environments may report immigration-related perceptions in ways that reflect narrative salience more than actual enforcement conditions. That creates the possibility that media-driven context effects enter survey responses indirectly, even when the administrative environment changes more gradually.

The practical implication, then, is not that narrative data are useless. Rather, narrative indicators may be useful as measures of public attention, issue framing, or anticipatory discourse, but they should not be treated as interchangeable with benchmark enforcement measures. In applied settings, this means they are better used as complementary indicators than as substitutes.

Conclusion

This study evaluates the relationship between immigration narratives and enforcement activity through a TSE framework. The evidence shows modest contemporaneous association, somewhat stronger lagged association, and limited explanatory power in a simple regression model. These results indicate that narrative indicators are related to enforcement activity, but only incompletely and not in a stable contemporaneous way.

The main methodological conclusion is that narrative indicators should be treated with caution when used as proxies for underlying enforcement conditions. The main substantive conclusion is that public narratives about immigration may move with enforcement activity, but often with lag, attenuation, or additional volatility introduced by media and platform dynamics.

Future work should go beyond contemporaneous correlation and simple linear regression by estimating richer time-series models, incorporating formal lag structures, and extending the benchmark set to include additional administrative and survey-based measures. That would help determine whether the observed misalignment is best understood as delay, nonlinear coupling, or broader measurement distortion.

References

Groves, R. M., Fowler, F. J., Couper, M. P., Lepkowski, J. M., Singer, E., & Tourangeau, R. (2009). *Survey Methodology*. Wiley.

Valliant, R., Dever, J. A., & Kreuter, F. (2013). *Practical Tools for Designing and Weighting Survey Samples*. Springer.

Lazer, D., Kennedy, R., King, G., & Vespignani, A. (2014). The parable of Google Flu: traps in big data analysis. *Science*, 343(6176), 1203–1205.

Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press.

Salganik, M. J. (2019). *Bit by Bit: Social Research in the Digital Age*. Princeton University Press.

Gentzkow, M., Kelly, B., & Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3), 535–574.

Mendelsohn, J., & Budak, C. (2025). When People Are Floods. arXiv.

Shiller, R. J. (2017). Narrative economics. *American Economic Review*, 107(4), 967–1004.

Shiller, R. J. (2020). *Narrative Economics: How Stories Go Viral and Drive Major*

Economic Events. Princeton University Press.

Appendix

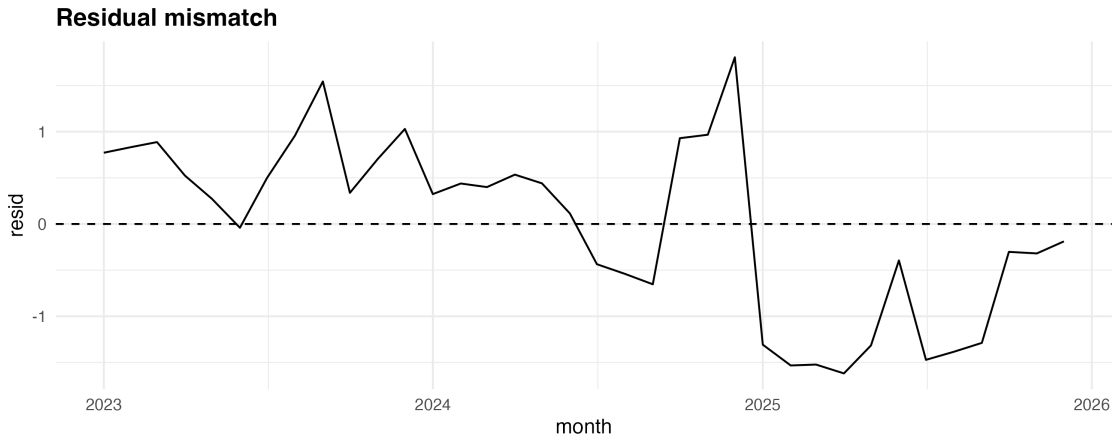

Figure 4: Residual Mismatch

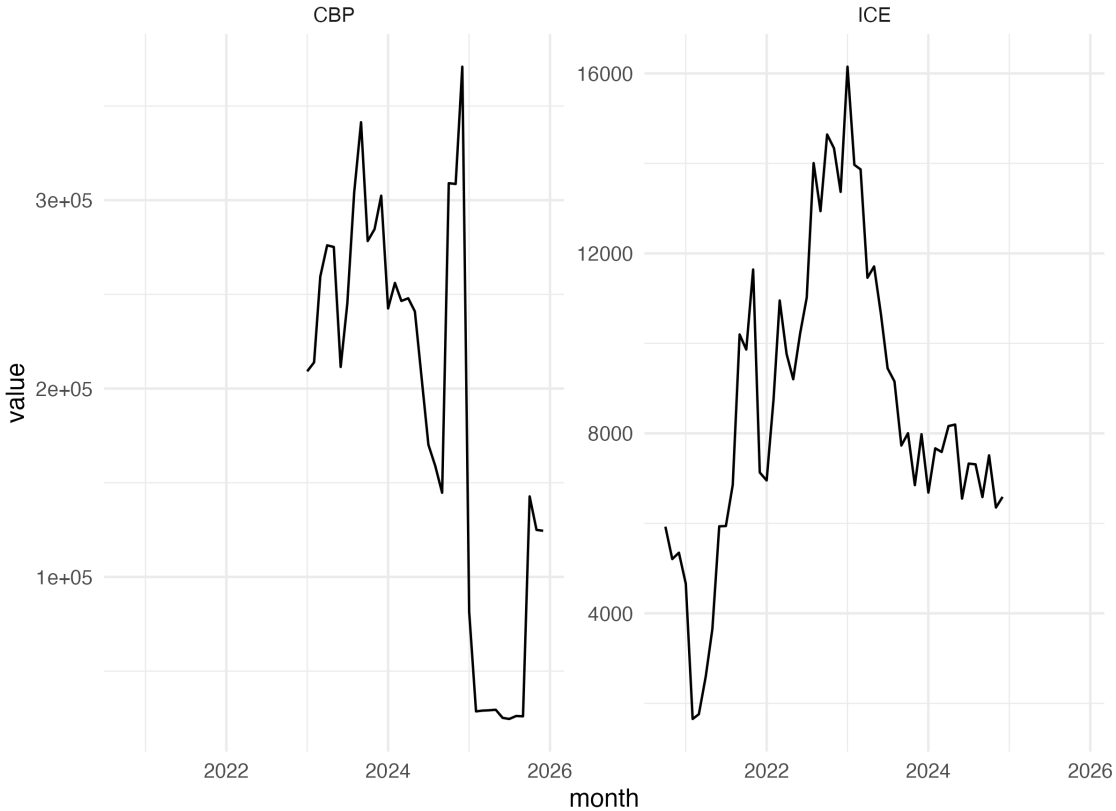

Figure 5: Raw Enforcement Trends

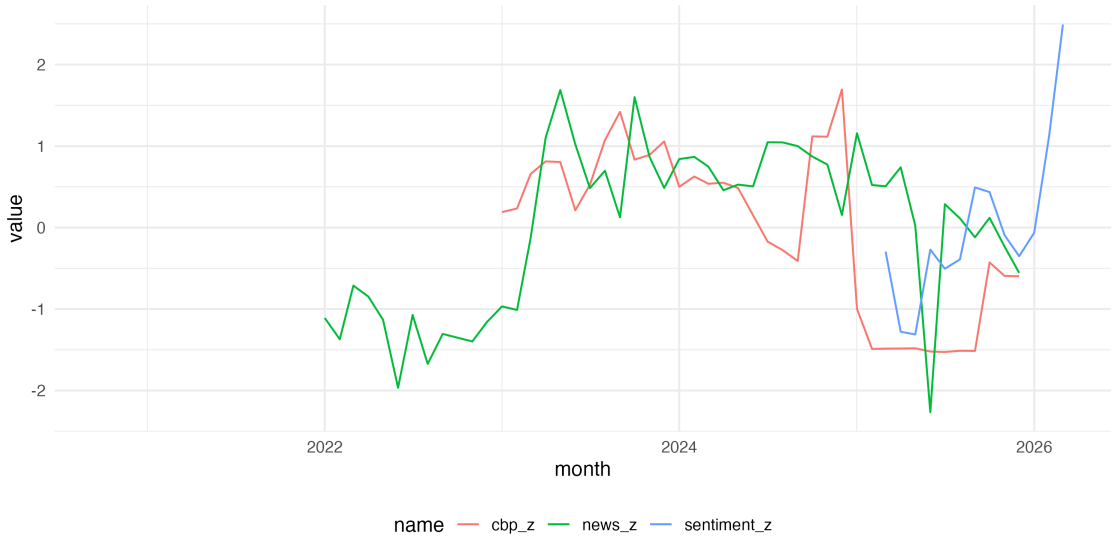

Figure 6: Narrative Comparison

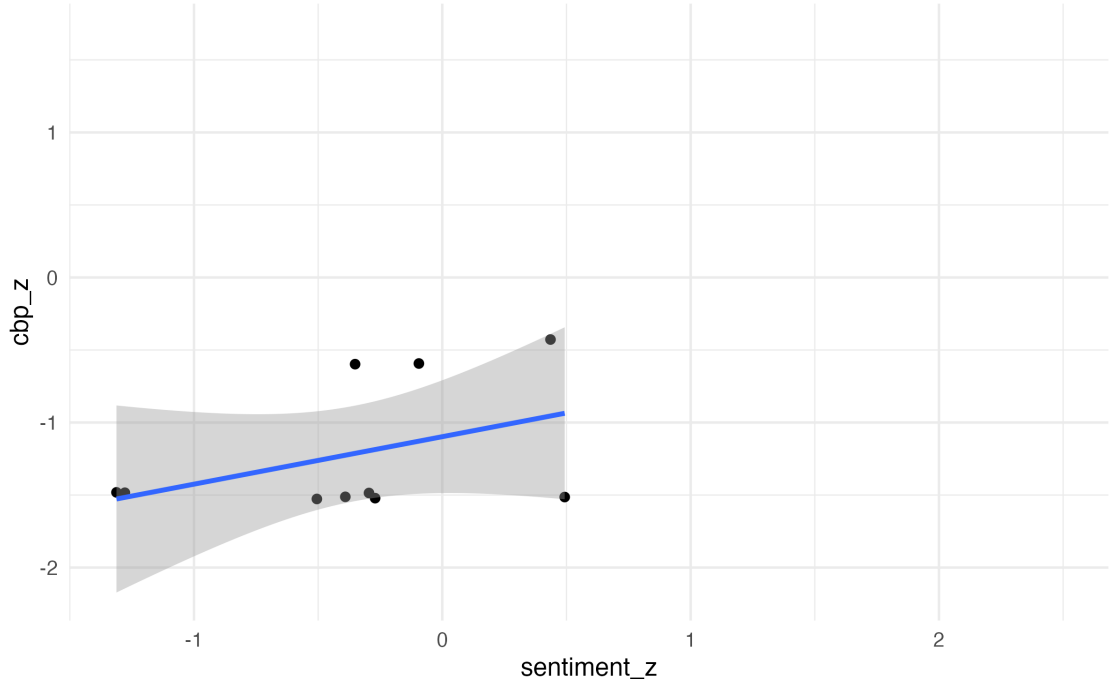

Figure 7: Sentiment vs. CBP Encounters

Measure	Mean
News	7464.458333333334
Tone	-2.5533835072306124
Reddit	1535.076923076923
ICE	8588.274509803921
CBP	188727.5555555556

Table 1: Summary Statistics

Measure	Months
News	48
CBP	36
ICE	51
Sentiment	13

Table 2: Overlap Between Narrative and Enforcement Signals